Lab Session 1: From the Unstructured Web to Structured Entities

Course: Web mining & semantics

Duration: 3 Hours (two sessions)

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1. Introduction

Les sites les plus simples à utiliser sont wikipédias

This combined session focuses on the first two steps of the knowledge engineering lifecycle: **Data Acquisition** and **Information Extraction**. You will transform raw, noisy HTML from the open web into a structured dataset of entities and relationships.

This is the "foundation" phase. The quality of the Knowledge Graph you build in later sessions depends entirely on the cleanliness of the data you extract today.

You should start thinking of the target site to crawl/scrap, but it is not necessary to set your target in this session. The objective of the sessions is to build a pipeline for information retrival, which you can use easily once the target sites are decided.

2. Learning Objectives

By the end of this lab, you will be able to:

- Build a **domain-specific crawler** that respects web ethics (`robots.txt`).
- Implement **boilerplate removal** to isolate meaningful content from UI noise.
- Apply **Natural Language Processing (NLP)** to identify key entities (NER).
- Perform **Dependency Parsing** to identify candidate relations between entities.

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3. Technical Requirements

Ensure your Python environment has the following libraries installed:

- `httpx` or `requests` (Fetching)
- `trafilatura` (Advanced text extraction)
- `spacy` (NLP pipeline)
- `pandas` (Data management)
- `en_core_web_trf` is a transformer-based model for NLP. it is equipped with these components: transformer, tagger, parser, ner, attribute_ruler, lemmatizer.

```
pip install trafilatura spacy pandas httpx
python -m spacy download en_core_web_trf
```

4. Phase 1: Web Crawling & Cleaning (90 Mins)

1.1 Source Selection

Choose a domain (e.g., *Sustainable Energy*, *Local History*, *news* or *AI Research*). Select 5–10 "seed" URLs that contain rich, text-heavy descriptions. You can also choose APIs as information sources, if you encounter the problem of anti-robotic checking. (You won't need `beautifulsoup` or `trafilatura`) if you extract text from APIs.

1.2 The Extraction Pipeline

Don't just scrape the whole page. A Knowledge Graph built on "Copyright 2024" or "Click here to subscribe" is useless.

- **Action**: Use trafilatura to fetch and strip HTML. It is specifically designed to find the "Main Content" of a page.
- **Storage**: Store your results in a JSONL (JSON Lines) format to preserve the mapping between the URL and the text.

Challenge: Write a function that checks if a page is "useful" (e.g., contains more than 500 words) before saving it.

5. Phase 2: Information Extraction (90 Mins)

2.1 Named Entity Recognition (NER)

Identify the "Nodes" of your future graph.

- **Action**: Process your cleaned text using spaCy.
- **Goal**: Extract labels such as PERSON, ORG (Organization), GPE (Location), and DATE.
- **Refinement**: Filter out common nouns. You only want specific instances that define your domain.

2.2 Introduction to Relations

A Knowledge Graph needs "Edges."

- **Action**: Look for entities that appear in the same sentence.
- **Experiment**: Use `spaCy` dependency tags (like `nsubj` and `dobj`) to see if you can programmatically find a verb connecting two entities.

Example: "The University of Paris [ORG] is located in France [GPE]."

Target Triple: (University of Paris) -> [is located in] -> (France)

6. Deliverables

You do not need to submit the results. The results will be used in next lab sessions, as part of the project. The results will consist of two files:

- crawler_output.jsonl: The cleaned text and metadata for your URLs.
- extracted_knowledge.csv: A table of entities found, their type, and the source URL.

Lab Report (2 pages max):

Methodology: How did you handle navigation? discribe the pipeline.

Ambiguity Analysis: Provide 3 examples of "Entity Ambiguity" (e.g., a word the AI got wrong due to context).

Scaling Reflection: How would your approach change if you had to crawl 10,000 pages instead of 10?